Agora: An LLM-augmented Boulware Agent for the ANAC 2026 HAN League

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Abstract

Agora is a hybrid agent for the ANAC 2026 Human-Agent Negotiation (HAN) League, designed to negotiate effectively against humans (and LLM stand-ins for humans) under the alternating-offers protocol augmented with free-text communication. This report documents the submitted agent, whose class is `agora.AgoraNegotiator`. Its core idea is to separate *decision making* from *post-decision wording*: a rule-based Boulware core decides every accept, reject and offer move from turn progress, the agent's own utility function, an estimated opponent model, and bounded text signals, while the generation layer wraps the already-decided move in persuasive, register-matched language. Side-channel LLM calls may seed the opponent and text state, but the free-form text generator can never rewrite the chosen move; if those calls time out, keyword and frequency fallbacks keep the strategy legal. We describe each component in detail (a blended time signal, a two-source opponent model, an opponent-aware multi-candidate proposal search ranked by an ASTRA-style acceptance-probability scorer, eight acceptance checks plus a final-step END guard, and a gated, self-remediating text layer), and report freshly measured offline results for this exact build: it finishes first in decision-only tournaments on the fixed HAN scenarios, reaches agreement in 94% of matches with zero crashes, matches the later (now retired) build's offline decision score to within 0.002, and ablations show the concession-rate-dependent proposal path and the ASTRA scorer each contribute measurable advantage. We close with implementation notes, stress-test results, limitations, and future work.

1 Introduction

The HAN League asks agents to negotiate with human partners, not other agents. Two features make this distinct from agent-vs-agent leagues. First, the channel is *multi-modal*: alongside a formal offer, each side may exchange free natural-language text used to persuade, explain, or mislead. Second, evaluation rewards the utility an agent secures, how human partners *perceive* it (measured through post-negotiation questionnaires), and how well it models its partner. In this setting, an agent needs a sound bargaining strategy, a credible text presence, and enough fault tolerance to survive real humans, slow infrastructure, and adversarial inputs without crashing or stalling.

Agora addresses these by keeping the bargaining policy rule-based and auditable, treating post-decision language as a separate, replaceable layer, and allowing partner text to influence the policy only through bounded state variables. This paper documents the agent's design and the reasoning behind each choice, and reports the experiments used to validate them.

2 Background and related work

Agora is built on the NegMAS platform [6] and its LLM-augmented negotiation tooling, extending a Boulware-based LLM negotiator. The underlying offering tactic is the classic time-based *Boulware* concession function [1], which holds out near its best outcome and concedes sharply only as the deadline nears; its counterpart Conceder and the Linear tactic anchor the baseline comparisons. For proposing, we draw on the Nash bargaining solution [3] as one of several candidate generators, and on the BOA (Bidding, Opponent-model, Acceptance) decomposition and frequency-based opponent models surveyed by Baarslag et al. [4]; our acceptance logic incorporates the "next-offer" acceptance condition (ACNext) [5]. The proposal scorer follows ASTRA [7], which ranks candidate offers by a saturating estimate of the opponent's acceptance probability rather than by raw value. For the text layer, the cost-aware gating of LLM calls is motivated by work on the price of LLM "thought" [8], and the self-rewriting safety pass follows assistive, socially-aware negotiation agents [9].

3 Architecture

Agora overrides four aspects of its base negotiator: how it responds to offers, how it proposes, the text prompts, and how it models the opponent. The design separates a bargaining policy from post-decision wording, with bounded side-channel text parsers feeding the policy state.

The **decision layer** is rule-based. Given the current turn progress, the agent's own utility function, an estimated opponent model, and bounded text-signal state, it produces exactly one action: accept, reject, or a specific counter-offer. It never asks the free-form generation layer what to do. Side-channel LLM calls are restricted to cold-read priority extraction and structured text parsing, and each has a keyword or frequency fallback, so language-model latency or failure degrades the state estimates rather than leaving the agent without a legal move.

The **post-decision generation layer** runs only after the move is decided. It wraps that move in natural language, mirrors the partner's tone and register, and is guarded throughout so that a slow or dead LLM backend falls back to deterministic templates without ever changing the decision.

Two cross-cutting safeguards protect the agent. First, local guards and fallback paths cover the common failure modes that would otherwise lose a match: malformed partner offers, missing utility functions, unavailable outcome tables, and LLM failures all route to a legal reject or Boulware-style move. Second, all incoming partner text is sanitised before it reaches the LLM or the keyword parser, neutralising a curated set of prompt-injection phrases (instruction-override and role-reset patterns) and stray markdown/tool-call markers, so the partner cannot redirect our own language model.

4 Time signals

Proposal generation reads a blended effective clock,

$$t_{\text{eff}} = \max\left(\tfrac{\text{step}+1}{n_{\text{steps}}},\ 0.5\,\tau_{\text{wall}}\right),$$

where τ_{wall} is the fraction of wall-clock time elapsed. Step-based progress normally dominates, which keeps the concession curve stable even when each LLM reply takes 20–60 s and would otherwise make the wall-clock race ahead of the discrete turns; the $0.5\,\tau_{\text{wall}}$ term still pulls the agent down its concession curve when a real time deadline is binding, so it never times out without having conceded. When proposing, the agent *hardens* this clock ($t_{\text{eff}} \leftarrow t_{\text{eff}}^{1.3}$ while the opponent is actively conceding) to stay tougher and capture more surplus.

Acceptance is deliberately asymmetric. In the shipped configuration it reads the raw wall-clock fraction τ_{wall} directly, which keeps the setting we validated against bot opponents; only if wall time runs far ahead of discrete progress does it fall back to the unhardened effective clock t_{eff}. Hardening the acceptance side was shown in tuning sweeps to reject marginal deals we could otherwise have closed. The exponent of the underlying Boulware tactic [1] is selected per scenario from the size and utility spread of the outcome space (base 3.0, clamped to $[2.0, 4.0]$): wider, richer outcome spaces give us room to hold out longer, while narrow ones call for softer concession.

5 Opponent model

For each issue we estimate how much the opponent cares about it (a normalised weight w) together with how frequently they propose each value. These come from two complementary sources.

Cold-read (from text). On the opening turn a single LLM call infers stated per-issue priorities w_{cold} from the partner's first message and offer. This is the only signal available before offers accumulate and lets the agent reason about the opponent immediately.

Revealed preference (from offers). Once offers start to accumulate, a frequency opponent model [4] measures revealed importance, but replaces the usual raw concentration ratio with a smoothed Dirichlet posterior estimate,

$$\text{importance}_i = \frac{\alpha_0 + \max_v c_{i,v}}{\alpha_0\, k_i + \sum_v c_{i,v}}, \qquad \alpha_0 = 0.5,$$

where $c_{i,v}$ counts how often value v was offered on issue i and k_i is the number of distinct values seen. The estimate is high when the partner repeatedly insists on one value and is robust to small-sample noise; an entropy-based variant was tried first and reverted after it flattened the importance signal.

Fusion. Once both are available they combine as $w = 0.3\,w_{\text{cold}} + 0.7\,w_{\text{freq}}$, renormalised to sum to one. The two sources are kept apart on purpose: cold-read captures *stated* priorities and the frequency model captures *revealed* ones, so a strategic partner who verbally misrepresents their priorities cannot mislead us through either channel alone.

Use and caching. The cached base estimate for an outcome o is

$$s_0(o) = \sum_i w_i \frac{c_{i,v_i}}{\sum_v c_{i,v}}.$$

For the text-priority issue set P, the runtime score is

$$s(o) = \text{clip}_{[0,1]}\left(s_0(o) + 0.15 \sum_{i \in P} \frac{c_{i,v_i}}{\sum_v c_{i,v}}\right),$$

so text adds a small unnormalised boost before the final $[0,1]$ clamp. Because this is evaluated thousands of times per match during proposal scoring, it is cached and keyed so the cache self-invalidates when new offers arrive. A *confidence gate* (three offers seen, or a successful cold-read plus at least one offer) prevents the agent from acting on a model that has no real signal yet.

Scoring-only estimate. For HAN's opponent-model and deception scores, Agora exposes a Dirichlet-smoothed linear-additive utility estimate built from the same frequencies. This lets the league score how well we model the partner without that estimate ever feeding a decision. For issues whose values cannot be enumerated (fully continuous domains) the current build simply omits the issue from this scoring estimate; a finite, observed-value fallback for such domains is identified as future work in Section 12.

6 Offering strategy

The Boulware tactic supplies a base offer whose utility u_{base} acts as a floor below which the agent will not concede. On the very first turn the agent instead offers the $\sim$ 99th-percentile outcome, giving up less than 0.01 of utility; this signals room to concede without materially weakening our position and improves how humans perceive the opening.

When the confidence gate is satisfied *and* the opponent is conceding (concession slope > 0.03), the agent constructs four candidate offers, all constrained to $u_{\text{us}} \geq u_{\text{base}}$ so we never sacrifice our own utility:

- a *band search* for the best-for-the-opponent outcome in the narrow window $[u_{\text{base}}, u_{\text{base}} + 0.03]$ above our floor;
- a *Nash* bargaining outcome [3] maximising the product $u_{\text{us}} \cdot u_{\text{opp}}$;
- a *Pareto walk* [2] that greedily improves our utility starting from the partner's last offer; and
- the best outcome on a precomputed *Pareto front*.

The conceding gate is kept because, in tuning against fixed-threshold opponents that cannot be "targeted" from frequency data, enabling opponent-aware proposing unconditionally reduced score.

The four candidates are ranked by an ASTRA scorer [7],

$$\text{score}(c) = \lambda_u\, u_{\text{us}}(c) + \lambda_a\, p_{\text{acc}}(c),$$
$$p_{\text{acc}}(c) = \sigma\big(k(u_{\text{opp}}(c) - m_\star) + b\big),$$

where $m_\star$ is the sliding-window maximum opponent-utility over the last ten offers we have shown the partner. The intuition is that the partner anchors on our recent generosity, so a new candidate's acceptance probability saturates in the *gap* over that reference rather than in its raw value; the logistic concentrates the decision signal near the threshold that actually matters. The defaults $(\lambda_u, \lambda_a, k, b) = (0.6, 0.4, 6.0, 0.0)$ are all tunable. The winning candidate is then refined by a *win-win swap* toward the partner's most frequent values, which permits a small, bounded give-back but never drops below reservation+0.02.

When no confident model exists, the agent simply ships the base Boulware offer, and, to avoid monotonously repeating the same offer, swaps in an equally-valued outcome it has not yet proposed.

7 Acceptance

Before considering acceptance, a separate final-step END guard lets the agent politely terminate only when the last turn is genuinely hopeless. Otherwise, rather than the stock Boulware rule, the agent accepts an offer of utility u when it clears a composite threshold assembled from eight considerations, evaluated in order:

1. *Early accept:* if the offer is excellent ($u > 0.85$) and we are early ($t < 0.3$).
2. *Aspiration:* otherwise the threshold begins at the Boulware level $1 - t^e$.
3. *Text nudge:* a bounded adjustment from text signals (urgency, threats, concessions).
4. *Reservation floor:* the threshold is never allowed below the walk-away value.
5. *Late-game relaxation:* the threshold eases near the deadline, except when the partner's text shows urgency without threats (a sign they are anxious to close), in which case we hold firm.
6. *ACNext* [5]: never reject an offer better than what we would propose next.
7. *Convergence patience:* if the opponent is steadily conceding and waiting is likely to pay off, the threshold is briefly raised.
8. *Best-offer floor:* before the very end, require at least 90% of the best offer seen so far.

The opponent's concession rate used in several of these stages is the slope of a short linear regression over their recent offers (seven samples, or three in very short negotiations), which is robust to offers that oscillate up and down.

8 Text layer

The text layer makes the agent's language persuasive and readable while keeping it cheap and safe. Its tone switches between a persuasive and a firmer "deception" style only after a sustained signal: five flat or non-conceding offers to become firmer, two conceding offers to return to persuasion, and in both directions a three-offer dwell guard, so the partner never experiences jarring back-and-forth shifts. An LLM personality classifier, run only in longer negotiations ($n_{\text{steps}} > 20$) and at most once every eight turns to protect the latency budget, labels the partner as collaborative, tough, manipulative, or erratic, and routes text hints accordingly; the generated text mirrors the partner's tone, length and register but never their hostility.

Two gates keep the LLM affordable and safe. A *substance gate*, motivated by cost-aware use of LLMs [8], sends a partner message to the expensive structured LLM parser only when it has at least six words and a meaningful trigger (an urgency, threat, concession or fairness keyword, a question mark, or a digit); routine replies such as "ok" or "sure" skip the LLM entirely and use a keyword parser, saving 25–70 s per turn on local hardware. A one-shot *remediator*, following socially-aware negotiation agents [9], runs a rewrite pass over *our own* draft only when it looks hostile or exceeds 250 characters, stripping aggression, manipulation, contradictions and hallucinated values while keeping issue names and the words yes/no/accept/reject exactly as written. Finally, if any LLM call times out, a template assembles coherent text at no cost (opener × middle × ender), so the agent always produces a legible message.

9 Implementation notes

Building a custom strategy on top of a meta-wrapper required one subtle fix worth noting for users of the same pattern. The league constructs the agent and passes the utility function as a constructor keyword; the wrapper forwards it only to the inner Boulware agent, so it must be explicitly re-attached to the outer agent after initialisation (together with the public `id`/`name`). Without this, every guard that checks for the utility function short-circuits and the custom acceptance, opponent-aware proposing, Nash/Pareto candidates and high-anchor opener silently fall back to plain Boulware, and the league `calc_scores` path records an Advantage of 0. With the utility function correctly attached the full strategy is active: on Grocery and Island (LLM disabled, league scoring path) Agora scores about 1.8–2.0 on normalised Grocery, beating the Boulware, Conceder and Linear baselines by 1.0 to 1.8 per match. We note for transparency that a controlled ablation isolating the post-baseline tuning and hardening layer from this constructor fix found that layer *score-neutral* against time-based reference agents: the constructor fix is the change that makes the custom strategy measurable, and the remaining hardening (sanitisation, crash guards, text–outcome alignment) improves the human-facing channel and survival on slow infrastructure without changing the tuned bargaining parameters.

10 Evaluation

10.1 Tournament performance

Table 1: Offline decision-only `cartesian_tournament` (LLM disabled, $n_{\text{steps}} = 30$) against NegMAS reference agents, freshly measured on this May build. Left: the three fixed HAN scenarios, 20 repetitions. Right: the same agents plus five seed-42 random scenarios, 10 repetitions. No crashes in either configuration. These reference-bot scores do not predict the live LLM/human evaluation.

Fixed HAN		Wider grid (+5 random)	
Agora	**0.689**	**Agora**	**0.672**
BOA	0.677	BOA	0.671
Boulware	0.677	Boulware	0.671
Linear	0.491	Linear	0.512
Conceder	0.283	Conceder	0.373

We ran two offline, decision-only tournament configurations (LLM disabled) against four reference strategies (Boulware, Linear, Conceder and a BOA agent). On the three fixed HAN scenarios (Grocery, Island, Trade), Agora finishes first by a clear +0.012 over the next agent (Table 1, left); on a wider grid that adds five randomly-generated scenarios it again finishes first but only edges the BOA and Boulware baselines by under 0.001, a statistical tie at the top (right). These reference agents are not the league's live LLM/human partners, so we read Table 1 as evidence of a sound decision core rather than of leaderboard standing; on the fixed scenarios, which are structurally closer to the league's human-designed domains than a random draw, the opponent-aware proposal ranking and the candidate generators add the measured lift over the Boulware base reflected in Table 2.

10.2 Agreement rate and self-play

In a decision-only head-to-head sweep (the three fixed scenarios $\times$ six opponents — a BOA agent, a second Agora in self-play, a simple frequency negotiator, Boulware, Linear and Conceder — $\times$ three repetitions, 54 matches, 100 steps, LLM disabled) the agent reaches agreement 94% of the time (51/54) with zero crashes. Per scenario the agreement rate is 100% on Grocery, 100% on Trade and 83% on Island; every no-deal is the same Island matchup against the intransigent BOA opponent that holds a fixed threshold the frequency model cannot target. In self-play the two Agora agents always agree (9/9), confirming the strategy does not deadlock against a copy of itself.

10.3 Parity with the later build

A direct, decision-relevant check supports the choice to reactivate this build rather than the later one. On the identical 54-match decision-only sweep, this May build reaches a composite Advantage-plus- Deception score of 1.563, statistically indistinguishable from the later June build's 1.561 on the same harness. The subsequent live-regime engineering therefore left the offline decision quality unchanged, so reverting to the build with a validated decision core costs nothing measurable in the bargaining regime while removing the unvalidated live-only changes.

10.4 Component ablation

Table 2: Contribution of decision-layer ablations, measured by re-running the 54-match decision-only sweep with one runtime monkey-patch at a time; the agent source is untouched. u_{norm} is normalised advantage in $[0, 1]$; Δ is relative to the full agent. Agreement is 94% in every row.

Configuration	u_{norm}	Δ
Full agent	0.648	—
− concession-rate proposal path	0.615	−0.033
− ASTRA acceptance-prob term	0.627	−0.021
− high-anchor opener	0.649	+0.001

The largest measured drop is the concession-rate-dependent proposal path (−0.033): forcing the concession-rate signal to zero closes the opponent-aware proposal gate and also removes proposal-time hardening, so this row measures that combined proposal behaviour rather than a pure gate-only isolation. The ASTRA acceptance-probability term is next (−0.021); both rows act entirely through advantage, leaving the deception estimate untouched, exactly as expected since they choose *which* offer to send and not how the opponent is modelled. The high-anchor opener has no measurable effect in this offline, bot-opponent regime (+0.001, within noise); by design its benefit lies in how a *human* perceives a generous opening, which this harness cannot score, so we keep it for the live rounds rather than for offline utility.

10.5 Parameter stability

The ASTRA scorer's two most sensitive knobs were swept jointly over a grid of twelve cells ($\lambda_a \in \{0.2, 0.3, 0.4, 0.5\} \times k \in \{4, 6, 8\}$), re-running the full 54-match sweep per cell. The grid shows little sensitivity: the agreement rate is a constant 94% in every cell and normalised advantage stays within $[0.645, 0.649]$, a spread of under 0.005. The shipped defaults $(\lambda_u, \lambda_a, k, b) = (0.6, 0.4, 6.0, 0.0)$ sit within 0.002 of the best cell, so they were chosen as a stable operating point rather than a sharp optimum.

10.6 Behaviour with the real LLM

With the real language model on short negotiations (the setting that matches the interactive GUI used for human play, $n_{\text{steps}} = 10$) the agent averages a utility of 0.817 per agreement. Instrumentation confirms that the post-decision wording layer does not rewrite offers or responses; differences between LLM-backed and decision-only runs come from the negotiation dynamics and the bounded side-channel text signals, not from the generator changing the selected action after the policy has chosen it.

10.7 Text-layer behavioural validation

Because HAN scores human perception in addition to utility, we validated the text gates directly, independently of the league score. On a talkative-opponent run the substance gate routed about 42% of messages to the cheap keyword path, comfortably above its $\geq 30\%$ design target, and in short interactive sessions it correctly made zero LLM calls when the human sent no text. The remediator fired on 47% of an adversarially-weighted test set (confirming the gate logic triggers when it should) but on 0% of messages under the production prompt, inside its $\leq 10\%$ design band. Both gates spend LLM budget only when it is warranted.

11 Robustness

Across more than 280 stress tests (extreme mechanism settings, pathological opponents, very long and prompt-injection text, invalid utilities, fresh installs on Python 3.13 and 3.14, tournament state-reuse, deterministic self-play, malformed partner offers, and a 200-scenario random-utility fuzzing sweep) we found and fixed five genuine bugs: a hang on outcome spaces with tens of thousands of options, the LLM occasionally hallucinating concrete offer values (fixed by instructing it to describe offers only qualitatively), a parallel-tournament hang from a misleading job-count default, an unsanitised injection pattern, and a crash on malformed opponent offers. Of 35 standard NegMAS negotiator classes, 34 are handled without crashing, the agent reaching agreement or correctly walking away; the only failure is an unimplemented method upstream. The agent also survives simulated LLM outages mid-negotiation, switching to templated text and still reaching a utility of 0.8. A cache of the opponent-utility computation, called on the order of 10^5 times per hundred matches, gave roughly a 5× speedup of the decision logic ($\approx 90 \rightarrow 17$ ms per match). The test suite comprises 157 unit tests covering the hardening helpers and decision-logic correctness.

12 Limitations and future work

The opponent model is a point estimate with only a coarse boolean confidence gate; it has no calibrated uncertainty, no temporal model of how an opponent's concession evolves, and no correlation structure across issues. A natural next step is a lightweight Bayesian posterior over opponent weights (for example a closed-form Dirichlet posterior, or a small particle filter), which would make the confidence gate principled and enable expected-utility proposal ranking instead of point-estimate ranking. The scoring-only opponent estimate could also be strengthened, both with an inverse-own-utility prior for the opposed-preference HAN scenarios and with a finite observed-value fallback so the deception score stays well-defined on fully continuous domains. The text layer is currently tuned and validated against time-based bots and LLM stand-ins; its persuasion and perception effects on real humans remain to be measured in the live rounds, and the personality classifier and tone policy could be retuned from that data. Finally, the strategy parameters were calibrated against the fixed HAN scenarios and standard reference agents; re-tuning against the live human-and-LLM distribution, once available, is left as future work.

13 Conclusion

Agora keeps post-decision text and strategy separated: a rule-based, auditable decision core, paired with a persuasion layer that adds tone mirroring and personality routing without rewriting the chosen offer or response. This separation lets the agent win a bilateral tournament against standard time-based competitors and continue operating under LLM failure or slow infrastructure, while keeping every move reproducible and leaving headroom for HAN's combined utility-and-perception scoring. The design is modular, so each component, most obviously the opponent model, can be strengthened independently as live human data becomes available.

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